

WEEK 2

```
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE BookFlow;
Database changed
mysql> CREATE TABLE Loans (
  ->   loan_id INT PRIMARY KEY,
  ->   member_id INT,
  ->   book_id INT,
  ->   loan_date DATE,
  ->   FOREIGN KEY (member_id) REFERENCES Members(member_id),
  ->   FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
  -> VALUES
  -> (1, 101, 1, '2026-07-20'),
  -> (2, 102, 2, '2026-07-21'),
  -> (3, 103, 3, '2026-07-22');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT Members.full_name, Books.title
  -> FROM Loans
  -> JOIN Members ON Loans.member_id = Members.member_id
  -> JOIN Books ON Loans.book_id = Books.book_id;

+-----+-----+
| full_name | title |
+-----+-----+
| Rahul Kumar | Harry Potter |
| Arjun Reddy | The Alchemist |
| Sneha Rao | Atomic Habits |
+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT published_year, COUNT(book_id) AS total_books
  -> FROM Books
  -> GROUP BY published_year;

+-----+-----+
| published_year | total_books |
+-----+-----+
| 1997 | 1 |
| 1988 | 1 |
| 2018 | 1 |
+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Donation_History (
  ->   donation_id INT PRIMARY KEY,
  ->   book_id INT,
  ->   donation_date DATE,
  ->   FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
  -> VALUES (4, 'Wings of Fire', 'ISBN004', 1999);
Query OK, 1 row affected (0.00 sec)
```

```

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Wings of Fire', 'ISBN004', 1999);
ERROR 1062 (23000): Duplicate entry '4' for key 'books.PRIMARY'
mysql>
mysql> INSERT INTO Donation_History (donation_id, book_id, donation_date)
-> VALUES (1, 4, '2026-07-30');
ERROR 1062 (23000): Duplicate entry '1' for key 'donation_history.PRIMARY'
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE INDEX idx_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SHOW INDEX FROM Books;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| books | 0 | PRIMARY | 1 | book_id | A | 3 | NULL | NULL | NULL | BTREE | | | YES | NULL |
| books | 0 | isbn | 1 | isbn | A | 3 | NULL | NULL | YES | BTREE | | | YES | NULL |
| books | 1 | idx_isbn | 1 | isbn | A | 4 | NULL | NULL | YES | BTREE | | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

```

```

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (5, 'The Great Gatsby', 'ISBN005', 1925);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Donation_History (donation_id, book_id, donation_date)
-> VALUES (2, 5, '2026-07-30');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title | isbn | published_year |
+-----+-----+-----+-----+
| 1 | Harry Potter | ISBN001 | 1997 |
| 2 | The Alchemist | ISBN002 | 1988 |
| 3 | Atomic Habits | ISBN003 | 2018 |
| 4 | Wings of Fire | ISBN004 | 1999 |
| 5 | The Great Gatsby | ISBN005 | 1925 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Donation_History;
+-----+-----+-----+
| donation_id | book_id | donation_date |
+-----+-----+-----+
| 1 | 4 | 2026-07-30 |
| 2 | 5 | 2026-07-30 |
+-----+-----+-----+
2 rows in set (0.00 sec)

```

